

ALIAH UNIVERSITY

Name of the Examination: Semester(II)

Session: 2021-22

Subject: Engineering Chemistry

Course code: CHMUGGE01

Duration: 3 Hrs

Full Marks: 80

(Use Separate Answer Sheet for Each Group)

Group A

1. Explain the following terms (*any TEN*): (i) State function (ii) Adiabatic process (iii) Molar heat capacity (iv) Reversible expansion (v) Enthalpy (vi) Intensive property (vi) Path function (vii) Internal energy (viii) Isochoric process (ix) Homegenous system (x) Isolated system (xi) Cyclic process 2.5x10

Answer any Three(3)

2. (a) Deduce the equation of work for reversible isothermal expansion of an ideal gas. (b) Prove that work done is more in reversible process than that of irreversible process. 2.5+2.5
3. (a) Establish the relation between ΔH and ΔU . (b) For the reaction $H_2F_2(g) \rightarrow H_2(g) + F_2(g)$. Calculate ΔH for the reaction. 2.5+2.5
4. (a) Derive $C_p = dq_p/dt$ and $C_v = dq_v/dt$. (b) Establish their relation for the ideal gas. 2.5+2.5
5. (a) What is Joule Thomson phenomenon? (b) Prove that for the Joule Thomson phenomenon, $\Delta H=0$. 2.5+2.5

Group B

Answer any Eight (8)

8x5= 40

6. Write short notes on:

- (a) Bakelite
- (b) Teflon

7. Give a brief account of:

- (a) Vulcanization of rubber
- (b) Polythene

8. Explain the following with suitable reactions

- (a) Phenol formaldehyde resin
- (b) Polystyrene

9. How are the following polymers prepared ? Mention their uses

- (a) PVC
- (b) LDPE

10. Write informative notes on the mechanism of radical polymerisation with suitable example

11. (a) What is meant by degree of polymerization ?

fruity
pan

(b) Distinguish between addition & condensation polymerisation with example

12. (a) How is the crepe rubber obtained from latex ?

✓ (b) Give the structural unit of gutta percha

13. Why are silicones called inorganic polymers? Discuss the synthetic procedure of linear chain silicones

14. Write note on conducting polymers with reference to oxidative or p-doping with mechanism

15. (a) How will you prepare Styrene-Butadiene Rubber (SBR). Mention its uses

(b) What is Ziegler-Natta catalyst ?

16. Write short notes on:

(a) Compounding of rubber

✓ (b) Conducting polyaniline

✓ 17. What do you mean by green chemistry. Give its importance

18. (a) What is meant by degree of polymerisation ?

✓ (b) How will you prepare polystyrene using anionic addition polymerisation method ?